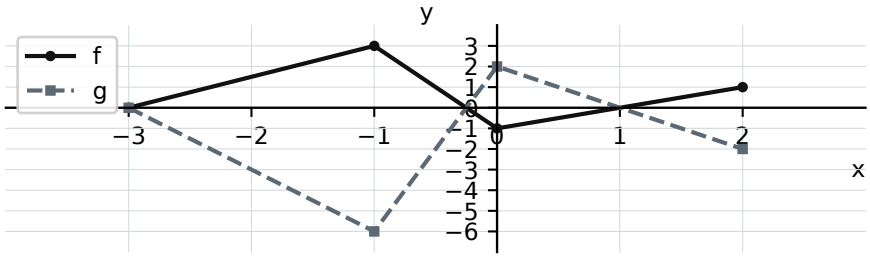
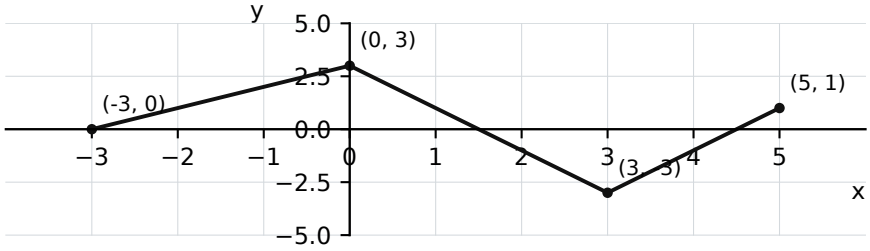
MATH 101 Unit 2: Functions and Their Graphs

Practice 2 | Version 202610122020.38 | 30 minutes

Name: _____ Date: _____

Complete and show all work on separate paper. Write exact final answers at right. Basic four-function calculator only.

QUESTIONS	ANSWERS
1. The solid graph is f and the dashed graph is g. Select the expression for g(x). $A.\quad g(x) = f(x) - 2$ $B.\quad g(x) = -2f(x)$ $C.\quad g(x) = 2f(x)$ $D.\quad g(x) = f(-2x)$ 	1. Choice 20 pts
2. From the graph, find (a) average rate from x = -3 to x = 5; (b) increasing and decreasing intervals; (c) the two local extrema (ordered pairs). 	2. (a) Rate 6 pts (b) Increasing 4 pts (b) Decreasing 4 pts (c) Local max 3 pts (c) Local min 3 pts
3. (a) Find the linear function through these points, with domain all real numbers. (b) Find its inverse and give the inverse domain and range. $(1,\ 3);\ (3,\ 9)$	3. (a) Function 10 pts (b) Inverse 6 pts (b) Domain 2 pts (b) Range 2 pts

MATH 101 Unit 2: Functions and Their Graphs

Practice 2 | Version 202610122020.38 | 30 minutes

Name: _____ Date: _____

Complete and show all work on separate paper. Write exact final answers at right. Basic four-function calculator only.

QUESTIONS	ANSWERS
4. For this function, find (a) f(-2), (b) domain, and (c) range. $f(x) = 3\|x - 2\| - 3$	4. (a) Value 6 pts (b) Domain 7 pts (c) Range 7 pts
5. (a) Choose the pair giving h(x) = f(g(x)). (b) Form F(G(x)) and evaluate it at the supplied input. $h(x) = (3x + 2)^2 + 4$ $A.\quad f(x) = x^2 + 4,\ g(x) = 3x + 2$ $B.\quad f(x) = 3x + 2,\ g(x) = x^2 + 4$ $C.\quad f(x) = x^2 + 2,\ g(x) = 3x + 4$ $D.\quad f(x) = 3x^2 + 4,\ g(x) = x + 2$ $(b)\quad F(x) = x^2,\ G(x) = x - 3,\ x_0 = 4$	5. (a) Choice 8 pts (b) Composition 8 pts (b) Value 4 pts

Instructor answer key: MATH 101 Unit 2: Functions and Their Graphs

Practice 2 | Version 202610122020.38 | 30 minutes

Each column matches the vertical answer positions on its corresponding test page.

Concise grading reference; required work and checks are assessed on separate paper.

TEST PAGE 1

1. Choice

B

20 pts

2. (a) Rate

$\frac{1}{8}$ 6 pts

(b) Increasing

$(-3, 0) \cup (3, 5)$ 4 pts

(b) Decreasing

$(0, 3)$ 4 pts

(c) Local max

$(0,\ 3)$ 3 pts

(c) Local min

$(3,\ -3)$ 3 pts

3. (a) Function

$f(x) = 3x$ 10 pts

(b) Inverse

$f^{-1}(x) = \frac{x}{3}$ 6 pts

(b) Domain

$(-\infty, \infty)$ 2 pts

(b) Range

$(-\infty, \infty)$ 2 pts

TEST PAGE 2

4. (a) Value

9 6 pts

(b) Domain

$(-\infty, \infty)$ 7 pts

(c) Range

$[-3, \infty)$ 7 pts

5. (a) Choice

A

8 pts

(b) Composition

$x^2 - 6x + 9$

8 pts

(b) Value

1

4 pts